# When Does Political Competition Stabilize Growth? Evidence from Developing Democracies*

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## Abstract

Growth volatility is a central constraint on economic development: it deepens poverty, shortens investment horizons, and destabilizes fragile democracies. Yet developing countries with similar macroeconomic fundamentals and exposure to external shocks often experience markedly different levels of growth instability. We argue that a key but neglected source of this variation is the structure of party competition. Investors in developing democracies face two distinct sources of policy uncertainty: how much policy will change across electoral outcomes, and how predictably a given government will implement policy once in power. The first increases with party system polarization; the second decreases. Their interaction produces a U-shaped relationship between programmatic differentiation and growth volatility. Using panel data from 16 Latin American democracies (1994–2017), we find that both the linear and squared terms of Dalton's party system polarization index are significant at the 1% level, with a turning point near 3.2 on a 0–10 scale. A mechanism investigation testing nine

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candidate channels finds no sequential mediator; the effect operates through a simultaneous shift in the aggregate expectations environment. The pattern replicates in a global sample of 68 developing and developed democracies before 2007 but attenuates afterward, consistent with a scope condition: when large external shocks dominate, domestic political signals lose their bite. These findings identify a political determinant of growth volatility that operates within democratic regimes, complementing existing work on macroeconomic fundamentals and institutional design.
*Keywords:* growth volatility; party system polarization; developing democracies; political economy of development; Latin America

## 1. Introduction

A common assumption in both academic and policy circles is that ideological moderation among political parties promotes macroeconomic stability. Convergent party systems produce smaller policy shifts after elections, which reduces uncertainty for investors and stabilizes growth. This assumption is wrong. In developing democracies, the most volatile growth environments are not those with the most polarized party systems. They are those where parties are ideologically *indistinct*. Moderate polarization stabilizes growth. It is convergence, not polarization, that is the underappreciated source of macroeconomic instability.

This matters because growth volatility is a central obstacle to economic development: it deepens poverty, shortens investment horizons, and destabilizes fragile democratic institutions (Mobarak, 2005; Loayza et al., 2007; Hnatkovska and Loayza, 2005). Yet developing countries with similar macroeconomic fundamentals experience markedly different levels of instability. Standard explanations (Easterly and Levine, 2001; Ramey and Ramey, 1995) leave much of this variation unexplained, and the growth volatility literature has almost entirely ignored how the structure of party competition shapes investor expectations.

This paper argues that investors face two types of policy uncertainty, not one. The first, how much policy will change across electoral outcomes, increases with polarization. The second, how predictably a given government will implement policy, *decreases* with it. Their interaction produces a U-shaped relationship between party system polarization and growth volatility. We formalize this through a variance decomposition of total policy uncertainty in Section 2.

Figure 1 documents the variation in growth volatility across Latin America. Figure 2 plots polarization against growth volatility: the binned conditional means trace the predicted U-shape, with a minimum near our estimated turning point of $P^* = 3.21$.

**Figure 1:** Variation in economic growth volatility across Latin America, 1993–2019. Panel A shows a country $\times$ year heat map (darker cells = higher volatility). Panel B plots country-level means $\pm 1$ SD, sorted by mean volatility. Panel C plots the regional annual mean with a $\pm$½ SD band, with the pre- and post-2007 period means indicated. Growth volatility = $|\Delta$real GDP growth rate$|$, year-on-year (World Bank WDI). 17 Latin American democracies.

**Figure 2:** Party system polarization in Latin America, 1993–2019. Panel A: regional mean Dalton vote index with 95% CI band and linear trend ($\beta = 0.067$/yr, $R^2 = 0.91$). Dashed coral line = estimated turning point $P^* = 3.21$. Panel B: country-level trajectories (coral = net increase $> 0.3$; teal = net decrease). Panel C: raw scatter of polarization against growth volatility with binned conditional mean (blue) and quadratic fit (dashed coral). Panel D: net change by country (late 2013–2019 minus early 1994–2005). Panel E: distribution of country-years across polarization levels over time. Polarization = Dalton (2008) vote-share weighted index (PELA surveys).

Using Dalton's (2008) vote-share weighted polarization index from elite legislative surveys (PELA), we estimate two-way fixed effects models in a panel of 16 Latin American developing democracies from 1994 to 2017. Both the linear and squared terms are significant at the 1% level. The turning point lies near 3.2 on a 0–10 scale, at the 75th percentile of observed polarization, meaning most country-years sit on the downward arm of the U where more polarization is associated with less volatility. Lagged specifications support a polarization-to-volatility direction. A global extension recovers the U-shape in 68 democracies before 2007. Nine candidate mechanisms produce null sequential mediation results; the effect operates through a simultaneous shift in the aggregate expectations environment. The relationship attenuates after 2007, consistent with a scope condition: when large external shocks dominate, domestic political signals lose their bite.

This paper contributes to the development literature in three ways. First, it identifies a political determinant of growth volatility that operates *within* democratic regimes, complementing existing work on macroeconomic fundamentals and regime type. Second, it shows that the effect of political competition on stability is nonlinear, overturning the conventional expectation that polarization uniformly increases volatility. Third, it identifies scope conditions under which political structures matter most, particularly in periods not dominated by large external shocks.

The paper proceeds as follows. Section 2 develops the theoretical framework. Section 3 describes the data and estimation strategy. Section 4 presents the main results. Section 5 examines robustness. Section 6 investigates mechanisms. Section 7 extends the analysis globally. Section 8 discusses development implications. Section 9 concludes.

## 2. Theoretical Framework

The existing literature on polarization and economic performance predicts a monotone positive relationship between polarization and growth volatility: as parties diverge, the expected magnitude of policy change following elections increases, investors discount long-term commitments, and volatility rises (Frye, 2002; Azzimonti and Talbert, 2014; Scartascini et al., 2013). Recent cross-national evidence confirms the negative linear relationship: Dorn et al. (2025) find that polarization

reduces growth through lower private investment, human capital investment, and total factor productivity in a panel of 75 countries, and Azzimonti (2018) shows that partisan conflict depresses private investment in the United States. Baker et al. (2020) show that polarized elections generate spikes in economic policy uncertainty, with downstream effects on firm behavior. This argument is intuitive and supported by growing evidence, but it captures only one of the two channels through which polarization affects investor uncertainty.

The missing channel concerns what the conventional argument holds constant. The standard framework focuses on the *between-regime* component of policy uncertainty, how much policy will differ depending on which party wins, but treats as fixed the *within-regime* component: how uncertain investors are about what a given party will actually do once in power. This paper builds on the existing framework by showing that within-regime uncertainty is itself a function of polarization, and that the two components move in opposite directions. Their interaction produces a nonlinear total effect on investor uncertainty. We develop this argument formally below.

### 2.1. The Investor's Problem: Policy Uncertainty as a Variance Decomposition

Consider an investor allocating capital across periods in a democratic political economy. The investor's problem is to forecast the policy environment $\tilde{p}$ (a vector including tax rates, regulatory intensity, trade policy, and property rights protection) over the relevant investment horizon. The realized policy environment depends on electoral outcomes that are uncertain at the time of investment. Formally, let $e \in \{L, R\}$ denote the electoral outcome, realized with probabilities $\Pr(e = R) = \pi$ and $\Pr(e = L) = 1 - \pi$, where $\pi$ reflects the electoral competitiveness of the contest. Conditional on the electoral outcome, the investor forms beliefs about realized policy: $\tilde{p} \mid (e = j) \sim F_j(\mu_j, \sigma_j^2)$ for $j \in \{L, R\}$, where $\mu_j$ is the expected policy under party $j$ and $\sigma_j^2$ is the conditional variance around that expectation.

The quantity that enters the investor's decision is the *total variance* of the policy environment:

$$\mathrm{Var}[\tilde{p}] = \underbrace{\pi(1-\pi)(\mu_R - \mu_L)^2}_{V_B\text{: between-regime variance}} + \underbrace{\pi\sigma_R^2 + (1-\pi)\sigma_L^2}_{V_W\text{: within-regime variance}} \tag{1}$$

This is the standard law of total variance applied to the mixture distribution of policies.[1] The between-regime component $V_B$ captures the standard policy risk argument: it increases in the squared ideological distance $(\mu_R - \mu_L)^2$ between the competing parties. The within-regime component $V_W$ captures something the conventional argument ignores: even knowing who wins, how predictable is policy? A party that wins the election may still govern in ways that deviate substantially from its stated positions , whether opportunistically or because coalition bargaining, institutional constraints, or interest-group pressure pull realized policy away from the electoral platform.

The conventional argument, in our framework, analyzes only $V_B$ and assumes $V_W$ is constant. We argue that this assumption fails systematically: $V_W$ is itself a decreasing function of party system polarization. The nonlinearity in total policy uncertainty follows as a consequence.

### 2.2. Polarization and the Two Components of Policy Uncertainty

**Between-regime variance.** Let party system polarization $P$ be proportional to the ideological distance between the parties: $P \propto |\mu_R - \mu_L|$.[2] Then $V_B = \pi(1-\pi)P^2$ is *increasing and convex* in $P$. This is the conventional result: as parties diverge, the spread of possible policy environments widens quadratically. The marginal contribution of additional polarization to between-regime variance is $\partial V_B / \partial P = 2\pi(1-\pi)P$, which is increasing in $P$. The between-regime channel, taken alone, always argues for less polarization.

**Within-regime variance.** The within-regime component $V_W$ is determined by how predictable each party is as a governing agent, which we argue is *decreasing in polarization*. The mechanism has three reinforcing channels.

*Reputational costs of deviation.* As parties occupy distinctive ideological positions, their electoral constituency becomes more concentrated and ideologically defined. Deviation from the

[1]The decomposition follows from $\mathrm{Var}[\tilde{p}] = \mathbb{E}[\mathrm{Var}[\tilde{p} \mid e]] + \mathrm{Var}[\mathbb{E}[\tilde{p} \mid e]]$, which yields equation (1) for a binary electoral outcome. The decomposition generalizes to multi-party systems with multiple electoral scenarios, with $V_B$ representing the variance of expected policies across scenarios weighted by their probabilities.

[2]This proportionality holds under Dalton's (2008) index, which calculates the vote-share weighted standard deviation of party positions. The specific formula is $PI = \sqrt{\sum_i p_i[(\bar{x}_i - \bar{x})/5]^2}$, which is increasing in pairwise ideological distances weighted by electoral relevance.

platform imposes larger reputational costs because the party's core voters, precisely those who chose it for its ideological identity, are the most sensitive to betrayal (Persson and Tabellini, 2000). Keefer (2007) shows that young democracies with non-programmatic parties cannot make credible policy commitments, leading to short-termist and clientelistic governance. A centrist party that promises vague moderation can implement almost any policy without alienating its median-voter base. A clearly left-wing party that nationalizes industries deviates only mildly from its supporters' expectations; a clearly left-wing party that privatizes massively faces a catastrophic credibility loss. Higher polarization therefore increases the cost of within-regime policy surprises, tightening the distribution of realized policy around the electoral platform.

*Organizational consistency.* Party systems with high programmatic differentiation produce parties with stronger internal discipline and more developed organizational infrastructure (Mainwaring and Scully, 1995; Kitschelt and Kselman, 2013). Party organizations built around ideological commitments generate consistent policy signals over time, selecting candidates with coherent platforms, building bureaucratic capacity around specific policy areas, and socializing members into shared policy frameworks. These organizational features reduce the variance of realized policy independently of any given election's outcome. Mainwaring (1998) documents precisely this pattern in Latin American party systems: institutionalized, programmatically differentiated parties deliver more consistent governance than clientelistic or personalistic ones, regardless of their left-right position.

*Investor learning.* When parties have clear and stable ideological identities, investors can update their beliefs about post-election policy more efficiently. Repeated interaction with a predictably left-wing or predictably right-wing party reduces Bayesian uncertainty about its policy intentions. Lupu and Riedl (2013) argue that political uncertainty in developing democracies is shaped precisely by the degree to which party systems provide stable, identifiable alternatives. Baker et al. (2016) show that policy uncertainty (as measured by their EPU index) is substantially higher when the composition of government is ideologically ambiguous. Our argument implies that the party system is an upstream determinant of the EPU: higher programmatic differentiation at the party system

level reduces the residual uncertainty that the EPU captures.

Formalizing these three channels: let $\sigma_j^2(P)$ denote the within-regime variance of party $j$'s policy as a function of system polarization. We assume $\partial\sigma_j^2/\partial P < 0$, with $\partial^2\sigma_j^2/\partial P^2 > 0$: within-regime variance is decreasing in polarization but at a diminishing rate, as programmatic clarity reaches a floor beyond which additional polarization cannot further reduce policy variance.[3] The within-regime component of total policy uncertainty therefore satisfies $\partial V_W/\partial P < 0$ and $\partial^2 V_W/\partial P^2 > 0$.

**The turning point.** Total policy uncertainty is:

$$\mathrm{Var}[\tilde{p}](P) = \underbrace{\pi(1-\pi)P^2}_{V_B(P)} + \underbrace{\pi\sigma_R^2(P) + (1-\pi)\sigma_L^2(P)}_{V_W(P)} \tag{2}$$

where $V_B$ is increasing and convex in $P$, and $V_W$ is decreasing and convex in $P$. At low polarization, $V_B \approx 0$ and $V_W$ is large; total uncertainty is high and dominated by within-regime unpredictability. As $P$ increases from zero, $V_B$ grows quadratically from near zero while $V_W$ falls, and $\partial V_W/\partial P$ is initially large in magnitude. For small $P$, the rate of decrease of $V_W$ exceeds the rate of increase of $V_B$, so total variance falls.

The turning point $P^*$ is defined by:

$$\left.\frac{\partial \mathrm{Var}[\tilde{p}]}{\partial P}\right|_{P=P^*} = 0 \quad \Longleftrightarrow \quad 2\pi(1-\pi)P^* = -\left.\frac{\partial V_W}{\partial P}\right|_{P=P^*} \tag{3}$$

At $P^*$, the marginal increase in between-regime variance exactly equals the marginal decrease in within-regime variance. Below $P^*$, total uncertainty falls with polarization; above $P^*$, it rises. This is the turning point of the U-shape, derived from the structure of the theory and not from the data. Our empirical estimate of $P^* \approx 3.2$ on the Dalton 0–10 scale quantifies this threshold in the Latin American institutional context.

[3] This follows naturally from the reputational cost argument: the marginal reduction in deviation risk from additional ideological differentiation diminishes as the party is already well-defined. Formally, $\sigma_j^2(P)$ can be modeled as a convex decreasing function such as $\sigma_j^2 = \bar{\sigma}^2 e^{-\alpha P}$ for some $\alpha > 0$, though the qualitative results do not depend on this functional form.

Proposition 1 formalizes:

> **Proposition 1.** Under the conditions described above, total policy uncertainty $\mathrm{Var}[\tilde{p}](P)$ is U-shaped in party system polarization $P$. There exists a unique interior turning point $P^* \in (0, P_{\max})$ below which total policy uncertainty is decreasing in $P$ and above which it is increasing in $P$.

Since growth volatility is increasing in policy uncertainty (Béjar and Mukherjee, 2011; Fatás and Mihov, 2003; Baker et al., 2016; Fernández-Villaverde et al., 2015), Proposition 1 implies a U-shaped relationship between party system polarization and growth volatility. This is our primary prediction:

> **Hypothesis 2 (H2):** *The relationship between party system polarization and growth volatility is curvilinear (U-shaped): growth volatility first decreases as polarization increases from low to moderate levels, then increases as polarization moves from moderate to high levels.*

H2 subsumes the conventional linear argument as a special case: over the upward arm of the U ($P > P^*$), the between-regime channel dominates and more polarization generates more volatility, consistent with Frye (2002) and related work. The theoretical contribution is identifying the downward arm, the region $P < P^*$ where the within-regime clarity effect dominates, and explaining why previous scholarship has missed it.

**Why previous studies found linear effects.** Three features of prior research produced linear estimates where the true relationship is nonlinear. First, samples restricted to high-polarization contexts (Frye, 2002) observe only the upward arm of the U, where the between-regime channel dominates. Second, polarization measures that conflate between-regime distance with within-regime clarity (including aggregate indices that assign high scores to both programmatically differentiated and chaotically fragmented systems) suppress the within-regime variance channel. The Dalton vote-share index, by measuring weighted ideological distance among electorally relevant parties, captures

the between-regime component while correlating with the within-regime component through its connection to party institutionalization. Third, the failure to separate the two channels analytically meant that no existing framework predicted a turning point, and linear specifications were imposed without theoretical justification for doing so.

### 2.3. Scope Conditions: When the Party System Signal Is Binding

The decomposition in equation (1) assumes that the party system is the primary source of policy uncertainty that investors face. This assumption is plausible under conditions of relative external stability, when the main source of uncertainty about the future policy environment is domestic political competition. It fails when large external shocks dominate investors' relevant information set.

**External shocks and signal dominance.** When commodity price booms, global financial crises, or sudden reversals in capital flows drive macroeconomic outcomes, the party system signal is swamped. Investors' relevant uncertainty is then about the external environment (commodity prices, global interest rates, risk appetite), not about which domestic party will win an election and what it will do. In Bayesian terms: the prior over external states has such high variance that the party system signal, even if highly informative about domestic policy, contributes negligibly to the posterior over macroeconomic outcomes. The mechanism through which polarization affects growth volatility is severed not because party system structure no longer shapes policy signals, but because policy signals no longer dominate the uncertainty that drives investment decisions and growth volatility.

This reasoning generates a third hypothesis:

> **Hypothesis 3 (H3):** *The U-shaped relationship between party system polarization and growth volatility attenuates when large external shocks (commodity price cycles, global financial crises) dominate domestic macroeconomic volatility.*

H3 is independently motivated by Raddatz (2007), who documents that external shocks are the primary driver of growth volatility in developing economies during periods of commodity and

financial turbulence, and by Funke et al. (2016), who show that financial crises generate polarization endogenously, further obscuring the causal ordering from party system structure to economic outcomes. The 2007–2008 global financial crisis and the subsequent collapse and reversal of the commodity super-cycle constitute the clearest instantiation of the conditions under which H3 predicts attenuation. The pre-2007 window provides the natural empirical test of H1 and H2 in the absence of dominant external shocks.

**Party institutionalization as a moderator.** A second scope condition follows from the within-regime variance channel. Recall that $\partial\sigma_j^2/\partial P < 0$: polarization reduces within-regime variance because clearer ideological positions generate reputational costs for deviation. But this effect is stronger when parties are *weakly* institutionalized at baseline. Where party institutionalization is already high, where parties have well-developed organizations, stable electoral bases, and strong programmatic commitments regardless of polarization level, the marginal benefit of additional polarization in reducing within-regime variance is small, because within-regime variance is already low. The turning point $P^*$ should therefore be *lower* in countries with more institutionalized party systems: less polarization is needed before the between-regime channel begins to dominate.

This prediction is not directly testable in our data given the limited within-country variation in party institutionalization, but it identifies a heterogeneous treatment effect that future research can examine in cross-national designs with larger samples.

### 2.4. Auxiliary Implications: Domestic versus Foreign Investors

The decomposition in equation (1) generates an additional prediction about the *differential* response of foreign and domestic investors that is directly testable in our data.

Foreign direct investors face a distinctive version of the investment problem. Unlike domestic investors, they operate across multiple country environments and can reallocate capital internationally. Their primary concern is not total policy uncertainty but the *tail risk* of adverse policy outcomes (nationalization, expropriation, discriminatory regulation) that are concentrated in the high-polarization, left-winning scenario. Because FDI involves long-horizon, relationship-specific

capital that is difficult to liquidate, foreign investors are particularly sensitive to the between-regime variance channel: they weight heavily the scenario in which a left-wing government dramatically shifts the regulatory environment. Kokashvili and Barbakadze (2020) find precisely this pattern: political polarization is associated with reduced FDI inflows, with the effect strongest where the left-wing electoral alternative has an explicitly nationalist or redistributive agenda.

Domestic investors, by contrast, cannot relocate easily and must therefore respond to *total* policy uncertainty, both the between-regime and within-regime components. Their investment decisions are sensitive to the overall variance of the policy environment, not just to the downside tail.

This differential response generates:

> **Proposition 2.** The stabilizing effect of moderate polarization on growth volatility operates primarily through the domestic investor channel. FDI volatility should exhibit a different pattern, potentially increasing with polarization over the range where domestic investor stabilization occurs, because foreign investors respond primarily to between-regime variance rather than total policy uncertainty.

Proposition 2 explains what might otherwise appear as an inconsistency in the mechanism analysis: the simultaneous observation of decreasing growth volatility and increasing FDI volatility at moderate polarization levels is not a failure of the theory but a prediction of it. It implies that the growth stabilization documented in our main results is driven by domestic investment behavior, a tightening of capital allocation decisions by firms and households that face the improved informational environment of moderate polarization, rather than by foreign capital inflows.

### 2.5. Hypotheses and Derivation Summary

The theoretical framework generates three testable hypotheses and two propositions. The first hypothesis reflects the conventional argument, nested as a special case:

> **Hypothesis 1 (H1):** *Higher levels of party system polarization increase growth volatility (the upward arm of the U-shape, valid for $P > P^*$).*

The primary prediction of the paper is:

> **Hypothesis 2 (H2):** *The relationship between party system polarization and growth volatility is U-shaped. Growth volatility first decreases as polarization increases from low to moderate levels (the downward arm, valid for $P < P^*$), then increases as polarization moves from moderate to high levels (the upward arm, valid for $P > P^*$). The turning point $P^*$ is defined by equation (3).*

The scope condition is:

> **Hypothesis 3 (H3):** *The U-shaped relationship between polarization and growth volatility attenuates in periods dominated by large external shocks, when domestic political signals no longer constitute the binding source of macroeconomic uncertainty.*

Propositions 1 and 2 follow from the assumptions of the framework. Proposition 1 establishes existence and uniqueness of the turning point $P^*$ under assumptions stated in subsection 2.2. Proposition 2 predicts divergence between domestic and foreign investor responses over the stabilizing arm of the U. Both propositions generate auxiliary implications that future research can test, including the role of party institutionalization as a moderator of $P^*$ (Proposition 1) and the domestic investment mechanism as the operative channel (Proposition 2).

The theoretical framework, in sum, differs from the conventional argument not by adding an additional linear channel but by respecifying the informational problem investors face. It treats total policy uncertainty, not merely between-regime distance, as the quantity that drives investment decisions and growth volatility, and shows that the two components of total policy uncertainty respond differently to party system polarization. The U-shape follows from this respecification under plausible conditions, not as an empirical observation seeking a theoretical home.

## 3. Data and Methodology

### 3.1. Dependent Variable: Growth Volatility

Our dependent variable is the absolute year-on-year change in the real GDP growth rate:

$$\text{Abs}(\Delta\text{growth})_{it} = |g_{it} - g_{it-1}|$$

where $g_{it}$ is the real GDP growth rate for country $i$ at time $t$. This measure accurately operationalizes temporal variance in growth within countries (Béjar and Mukherjee, 2011; Mobarak, 2005; Denizer et al., 2002) and avoids the biases of cross-sectional measures, which underestimate time-series variance and are sensitive to outliers. Data are from the World Bank Development Indicators (WDI). We use the natural logarithm of this measure in all models to address right-skewness.

### 3.2. Main Independent Variable: Party System Polarization

Our primary polarization measure is Dalton's (2008) vote-share weighted ideological polarization index:

$$PI = \sqrt{\sum_{i=1}^{n} p_i \cdot \left(\frac{\bar{x}_i - \bar{x}}{5}\right)^2}$$

where $p_i$ is the vote share of party $i$, $\bar{x}_i$ is its position on the left-right scale, and $\bar{x}$ is the weighted mean across all parties. The index ranges from 0 (complete convergence) to 10 (maximum dispersion). We construct this index from the Programa de Elites Parlamentarias de América Latina (PELA), which surveys legislators on their party's left-right self-placement. The Dalton vote-share index correlates strongly with the Taylor-Herman index ($r = .876$) and Huber's index ($r = .894$). We also report results using Taylor-Herman seat-share and vote-share alternatives as robustness checks.

### 3.3. Sample and Controls

Our primary Latin American sample comprises 16 democracies from 1994 to 2017, restricted to country-years with a Polyarchy score $\geq 0.5$ on V-Dem for which PELA data are available. The effective panel contains 292 country-year observations. Controls follow the growth volatility

literature (Easterly and Levine, 2001; Béjar and Mukherjee, 2011; Mathonnat and Minea, 2018): capital account openness (Chinn and Ito, 2006), the liberal democracy index (V-Dem), log inflation, government spending volatility, fixed capital formation volatility, an election year dummy, mean district magnitude, and the effective number of legislative parties (Schmidt, 2002). We include the lagged dependent variable to account for temporal dependence in growth volatility.

Two measurement concerns deserve attention. First, the Dalton index is constructed from PELA surveys of legislators, not from expert codings or manifesto content analysis. Legislative self-placement introduces measurement error: legislators may strategically misrepresent their party's position, and small survey samples in some country-rounds raise concerns about precision. Classical measurement error in the independent variable would attenuate our estimates toward zero, making our significant results conservative. Non-classical error (if legislators in volatile economies systematically misreport positions) could introduce bias, though we are not aware of evidence for this pattern in the PELA literature. Second, the effective number of parties (ENP) and party system polarization are conceptually related: fragmented systems can be either polarized or convergent. Including both as regressors risks partial collinearity. We retain ENP because it controls for the mechanical relationship between fragmentation and volatility, but we note that our results survive dropping ENP from the specification.

### 3.4. Estimation and Identification

Our baseline specification is:

$$\log(\text{Abs}(\Delta g))_{it} = \alpha + \beta_1 \text{Pol}_{it} + \beta_2 \text{Pol}_{it}^2 + \gamma \mathbf{X}_{it} + \lambda_i + \delta_t + \varepsilon_{it}$$

where $\lambda_i$ are country fixed effects and $\delta_t$ are year fixed effects. A U-shaped relationship (H2) implies $\hat{\beta}_1 < 0$ and $\hat{\beta}_2 > 0$, with turning point at $-\hat{\beta}_1/(2\hat{\beta}_2)$. We estimate all models using two-way fixed effects with standard errors clustered at the country level.

**Identification.** Country fixed effects absorb all time-invariant confounders; year fixed effects absorb common time shocks. The primary residual concern is contemporaneous reverse causality: volatile economies may generate polarized party systems. We address this through three strategies:

(i) *Lagged specifications.* We estimate models at $t-1$ and $t-2$ lags of polarization. If reverse causality dominates, the contemporaneous specification should be strongest. We find that the $t-1$ specification recovers significance for the squared term while the $t-3$ lag fades, a pattern inconsistent with contemporaneous reverse causality and consistent with a 1–2 year institutional transmission lag.

(ii) *Period split.* We show that the relationship holds pre-2007 but attenuates post-2007. Since financial crises generate polarization endogenously (Funke et al., 2016), the pre-2007 window provides a cleaner identification environment. If the pre-2007 finding were reverse-causal, the attenuation of the relationship in the more volatile post-2007 period should strengthen it, not weaken it.

(iii) *Out-of-sample replication.* We replicate the main finding in 62 non-Latin American democracies, ruling out unobserved regional confounders specific to Latin America as the driver of the relationship.

This package does not achieve the causal identification of a natural experiment or credible instrument. The evidence is consistent with the causal interpretation but does not definitively establish it.

# 4. Main Results

Figure 3 presents the fitted partial-effect curves for our five main specifications, with 95% confidence intervals on the marginal effects. The coefficient summary appears in Table 1.

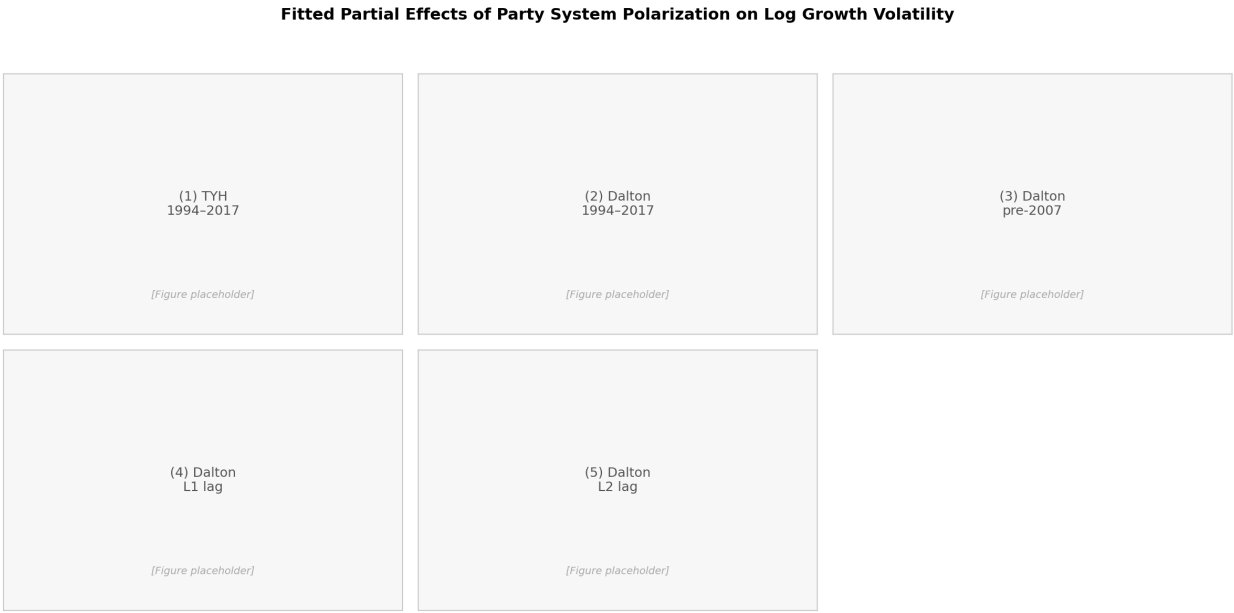


**Figure 3:** Fitted partial effects of party system polarization on log growth volatility across main specifications. Shading = 95% CI on the marginal effect. Histogram = empirical distribution of polarization in the sample. DV = $\log|\Delta\text{GDP growth}|$. Two-way FE, entity-clustered SE.

**Table 1:** Party System Polarization and Growth Volatility, Latin America

| | (1) TYH 1994–2017 | (2) Dalton 1994–2017 | (3) Dalton pre-2007 | (4) Dalton L1 lag | (5) Dalton L2 lag |
|---|---|---|---|---|---|
| Polarization ($\hat{\beta}_1$) | $-0.181$ | $-0.660^{***}$ | $-0.893^{***}$ | $-0.313$ | $-0.019$ |
| | (0.213) | (0.135) | (0.255) | (0.196) | (0.142) |
| Polarization$^2$ ($\hat{\beta}_2$) | 0.021 | $0.103^{***}$ | $0.136^{***}$ | $0.026^{**}$ | 0.001 |
| | (0.013) | (0.021) | (0.035) | (0.013) | (0.008) |
| Lagged DV | 0.004 | 0.007 | $-0.090^{*}$ | 0.013 | 0.007 |
| | (0.045) | (0.047) | (0.048) | (0.046) | (0.047) |
| FCF volatility | $0.021^{**}$ | $0.019^{**}$ | $0.032^{***}$ | $0.020^{**}$ | $0.021^{**}$ |
| | (0.008) | (0.008) | (0.010) | (0.008) | (0.008) |
| District magnitude | $-0.006^{***}$ | $-0.006^{***}$ | $-0.002$ | $-0.006^{***}$ | $-0.006^{***}$ |
| | (0.002) | (0.002) | (0.004) | (0.002) | (0.002) |
| Eff. N parties (ENP) | $-0.121^{*}$ | $-0.109^{*}$ | $-0.318^{***}$ | $-0.123^{*}$ | $-0.110$ |
| | (0.062) | (0.064) | (0.115) | (0.063) | (0.068) |
| Turning point | 4.27 | 3.21 | 3.28 | 6.01 | — |
| Shape | U | U | U | U | — |
| Controls | Yes | Yes | Yes | Yes | Yes |
| Country FE | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes |
| $N$ | 292 | 293 | 166 | 278 | 264 |
| $R^2$ (within) | 0.061 | 0.054 | 0.081 | 0.063 | 0.028 |

*Notes:* DV = log |ΔGDP growth (WDI)|. Controls include capital account openness, liberal democracy index, log inflation, spending volatility, FCF volatility, election year dummy, district magnitude, and effective number of legislative parties. Standard errors clustered at country level in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

The results provide strong support for H2. Model (1) uses the Taylor-Herman seat-share index; neither term achieves statistical significance over the full period. Model (2) uses Dalton's vote-share index, the measure directly motivated by PELA legislative surveys and the paper's theoretical focus on programmatic differentiation. Both the linear term ($\hat{\beta}_1 = -0.660$, $p < 0.01$) and squared term ($\hat{\beta}_2 = 0.103$, $p < 0.01$) are significant at the 1% level, with a turning point at 3.21 on the 0–10 scale. Model (3) restricts to the pre-2007 period; both terms remain significant at 1% with a stable turning point at 3.28.

The measure matters. The Dalton vote-share index substantially outperforms the seat-share alternative because it directly operationalizes what the theoretical mechanism requires: the weighted ideological distance among parties as voters, not legislatures, express it. Seat-share indices give greater weight to small parties that win disproportionate legislative representation under PR rules, potentially overstating the programmatic content of their presence. Vote shares better capture the informational environment investors observe.

Models (4) and (5) present lagged specifications. When polarization is measured at $t-1$, the squared term is significant ($p < 0.05$); at $t-2$, significance fades. This decay pattern is inconsistent with contemporaneous reverse causality: were volatile economies generating polarized party systems, we would observe a monotone decay from $t = 0$ outward. Instead, the $t-1$ specification recovers significance relative to the contemporaneous baseline, and then fades at $t-2$, a pattern consistent with a 1-year institutional transmission lag from party system configuration to investor behavior to macroeconomic outcomes.

Substantively, the turning point at 3.21 falls at approximately the 75th percentile of the observed polarization distribution (mean = 2.6, 75th percentile $\approx$ 3.4). Most Latin American country-years in our sample therefore operate on the downward arm of the U, the region where more polarization is associated with less volatility. The conventional linear prediction that polarization monotonically increases volatility does not describe the empirically dominant regime in the region.

# 5. Robustness

## 5.1. Period Heterogeneity

Figure 4 presents split-sample results and the country-level decomposition of changes in growth volatility across periods.

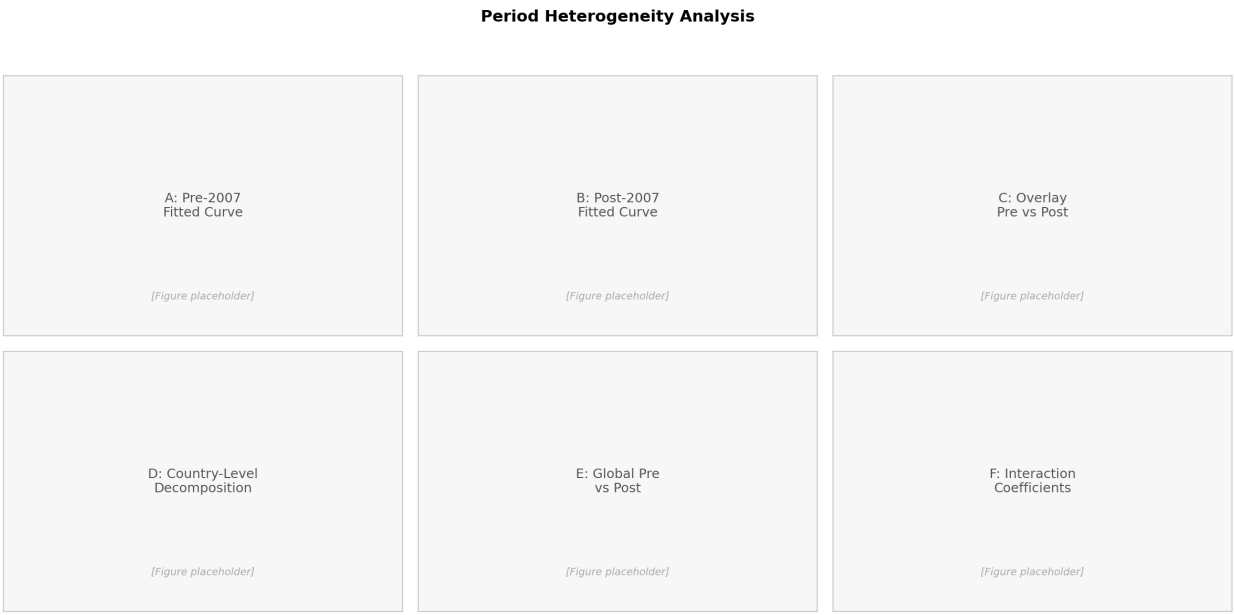


**Figure 4:** Period heterogeneity analysis. Panels A and B show fitted partial-effect curves for pre-2007 and post-2007 sub-samples. Panel C overlays the two fitted curves. Panel D decomposes country-level changes in mean growth volatility across periods. Panel E plots pre- vs. post-2007 fitted curves for the global sample. Panel F reports interaction model coefficients $\pm$ 95% CI.

In the pre-2007 Latin American sample, both the linear ($p < 0.10$) and squared ($p < 0.05$) terms are significant. In the post-2007 sample, neither term is significant. Panel D reveals why: the post-2007 period scrambles the regional pattern in ways orthogonal to domestic party systems. Uruguay and Venezuela experience dramatic volatility reductions driven by commodity booms and economic collapse respectively; Paraguay and Argentina become more volatile for reasons unrelated to party system configuration. The polarization signal cannot compete with these shocks.

The post-2007 attenuation is informative about causal direction. If volatile economies were causing polarized party systems (reverse causality), the relationship should strengthen in the more volatile post-2007 period, not weaken. The opposite pattern supports the scope condition: the

domestic political channel operates when external stability allows party system signals to bind.

### 5.2. Lagged Specifications across Samples

Figure 5 presents the lag structure systematically across the Latin American and global samples.

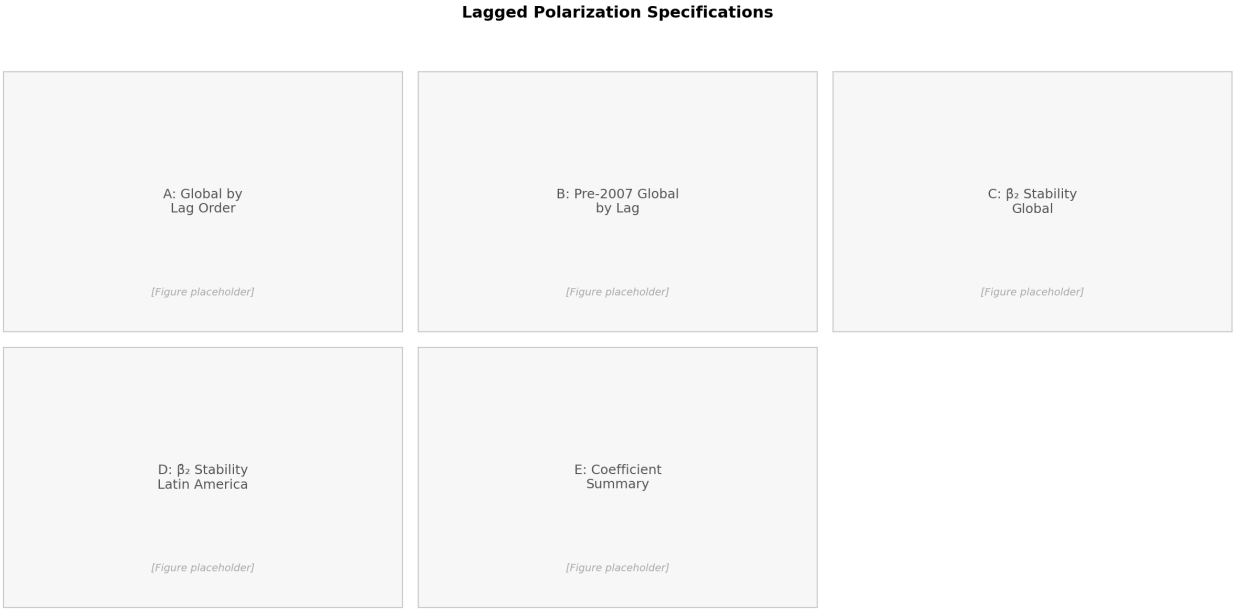


**Figure 5:** Lagged polarization specifications. Panel A shows fitted curves by lag order for global democracies. Panel B shows pre-2007 global results by lag. Panel C plots $\hat{\beta}_2$ stability across lag orders $\pm$ 95% CI for both the full period (blue) and pre-2007 (purple) global samples. Panel D shows the equivalent for Latin America (coral). Panel E summarizes coefficients across all specifications.

In the global pre-2007 sample, the squared term strengthens from $L0$ ($p < 0.01$) to $L1$ ($p < 0.01$) to $L2$ ($p < 0.01$) before fading at $L3$. In the Latin American sample, $L1$ recovers significance for the squared term relative to the contemporaneous baseline before fading at $L2$. Both patterns are inconsistent with contemporaneous reverse causality.

## 6. Mechanism Analysis

Figure 6 presents the results of the mechanism investigation.

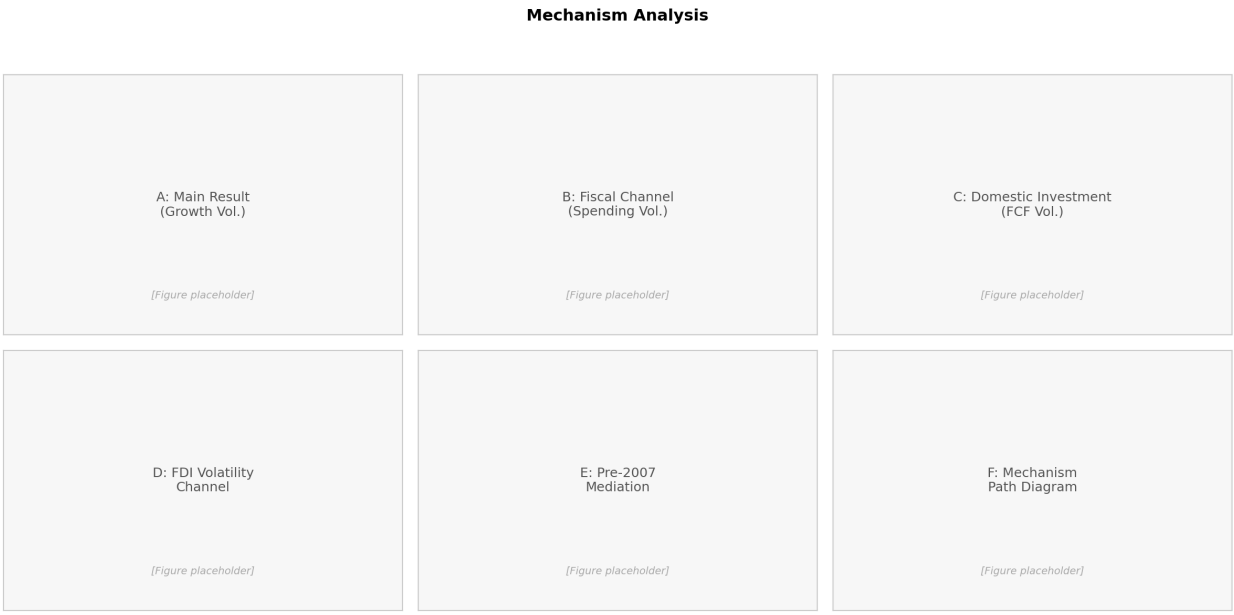


**Figure 6:** Mechanism analysis. Panel A reproduces the main growth volatility result (U-shaped, Dalton vote). Panels B and C test the fiscal channel (spending volatility) and domestic investment channel (FCF volatility). Panel D tests the FDI volatility channel. Panel E shows the pre-2007 mediation. Panel F presents the mechanism path diagram.

We submit every credible annual intermediate variable to the same two-way fixed effects specification used in the main analysis, testing whether it (a) displays a quadratic relationship with polarization and (b) mediates the polarization–volatility link sequentially. The candidate channels are: domestic investment volatility (gross fixed capital formation), FDI volatility, monetary volatility (M2 growth), trade openness volatility, government spending volatility, inflation volatility, social polarization, party institutionalization, and policy signal stability.

The results are uniformly null as sequential mediators. No intermediate variable attenuates the direct effect of polarization on growth volatility when included in the model. The fiscal channel, through which Béjar and Mukherjee (2011) showed electoral system effects on spending discretion produce growth volatility, does not operate here: polarization does not predict spending volatility, and adding spending volatility as a mediator produces zero attenuation of the polarization coefficients. The domestic investment channel (FCF volatility) and monetary channel (M2) show no quadratic relationship with polarization. Trade openness volatility is similarly null.

The FDI channel produces a more nuanced but ultimately non-mediating pattern. Polarization

exhibits an inverted-U relationship with FDI volatility ($\hat{\beta}_1 > 0$, $p < 0.10$; $\hat{\beta}_2 < 0$, $p < 0.05$), with a turning point near 6.1, substantially higher than the growth volatility turning point of 3.2. The two curves run in opposite directions in the empirically relevant polarization range: as polarization increases from low to moderate levels, FDI volatility rises while growth volatility falls. This divergence is consistent with Proposition 2: foreign investors, who can exit, respond primarily to between-regime variance (Kokashvili and Barbakadze, 2020), while the growth-stabilizing effect operates through domestic channels that respond to total policy uncertainty. At the firm level, Zhu (2024) shows that a one-standard-deviation increase in political polarization reduces firm investment by 1% in the United States, confirming that polarization affects real investment decisions, though the effect in that context is linear rather than U-shaped.

The one partial exception is inflation volatility, which displays the same U-shape as growth volatility with a similar turning point. But inflation volatility does not itself predict growth volatility in the model, so it functions as a co-outcome of the same underlying process rather than a mediator.

These null results are informative. The variance decomposition theory in Section 2 predicts that polarization reshapes the *entire* expectations environment simultaneously: both between-regime and within-regime variance shift together as polarization changes, affecting all investment margins at once. This is a diffuse, simultaneous mechanism, not a sequential one that routes through a single policy instrument. Finding that no individual channel mediates the effect, while the direct relationship holds across every specification, is precisely the empirical signature of a simultaneous expectations shift. The mechanism operates at the level of the aggregate uncertainty environment, and no single downstream variable captures enough of that shift to attenuate the direct effect.

We treat this as an honest account of the evidence. Direct measurement of the aggregate expectations environment, analogous to the Baker et al. (2016) EPU index but constructed for Latin American countries, would allow a more targeted test. Until such data are available, the null mediation results across all nine channels narrow the theoretical interpretation: whatever mechanism links polarization to growth volatility, it is not fiscal, not trade-mediated, not monetary, and not channeled through any single investment margin.

## 7. Global Extension

Figure 7 presents results from the global extension.

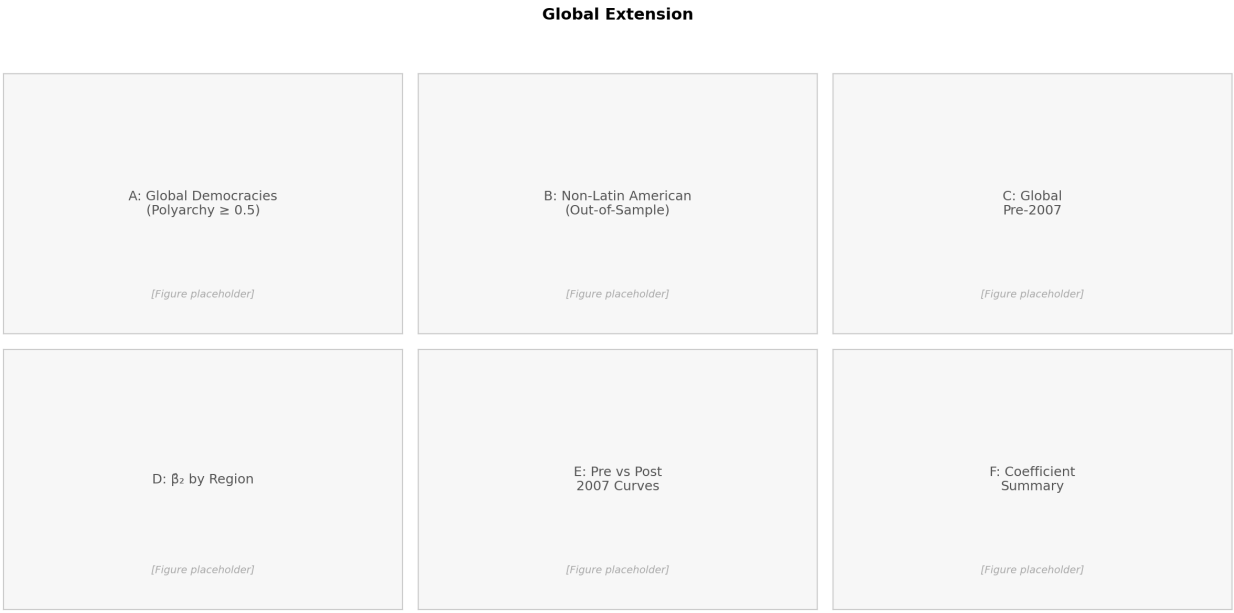


**Figure 7:** Global extension. Panel A shows results for global democracies (Polyarchy $\geq 0.5$), 1972–2020. Panel B shows non-Latin American democracies (out-of-sample test). Panel C shows global pre-2007 results. Panel D plots $\hat{\beta}_2$ by region. Panel E overlays pre- vs. post-2007 fitted curves. Panel F summarizes coefficients across samples.

We extend the analysis to a global sample using V-DEM's *v2cacamps* measure of political polarization, which captures societal division into political camps. This measure is conceptually distinct from Dalton's index of party-level ideological distance: *v2cacamps* is an expert survey indicator of societal polarization, while Dalton measures weighted ideological distance among electorally relevant parties. The two constructs are related (programmatic differentiation at the party level tends to produce societal division into camps), but they are not identical. We treat the global exercise accordingly: as an external plausibility check on whether the nonlinear pattern generalizes beyond Latin America and beyond a single operationalization, not as a direct replication.

The global pre-2007 sample (68 democracies, 1,207 observations) yields both the linear ($p < 0.05$) and squared ($p < 0.01$) terms significant, with a turning point near 3.67 on the rescaled index. The out-of-sample test in 62 non-Latin American democracies (Panel B) recovers the U-shaped

pattern ($\hat{\beta}_2$, $p < 0.10$), indicating that the finding is not driven by region-specific Latin American confounders. The post-2007 attenuation replicates globally (Panel E), consistent with the scope condition.

These results carry less inferential weight than the Latin American analysis. The polarization measure differs, the full global sample produces weaker findings, and the results are conditional on the pre-2007 period. We treat Latin America as the primary evidentiary base and the global results as evidence that the nonlinear pattern is not an artifact of one region or one operationalization, a useful finding, but not a substitute for direct replication with comparable measures.

## 8. Development Implications

The findings carry direct implications for how growth volatility affects development outcomes and for what policy levers are available to reduce it.

Growth volatility imposes costs that fall unevenly across the income distribution. Poor households, lacking savings and access to credit markets, cannot smooth consumption across downturns. Repeated growth fluctuations therefore deepen poverty traps: households that deplete assets during contractions start the next expansion from a lower base, generating persistent inequality even when average growth rates are positive (Loayza et al., 2007; Hnatkovska and Loayza, 2005; Malik and Temple, 2009). These effects are not transitory: Cerra and Saxena (2008) show that output contractions in developing economies produce permanent income losses, compounding the welfare costs of repeated fluctuations. Volatility also tightens credit constraints at the firm level. Lenders facing uncertain repayment environments raise risk premia or ration credit, and small and medium enterprises in developing economies bear the brunt (Aminou, 2024). The result is shorter investment horizons, reduced capital accumulation, and lower long-run growth. Ramey and Ramey (1995) estimate that a one-standard-deviation increase in growth volatility is associated with a 0.5 percentage point reduction in average annual growth, an effect large enough to compound into substantial welfare losses over a generation.

If party system polarization shapes growth volatility through the expectations channel our theory

proposes, then the structure of party competition is itself a development variable. This reframes a set of institutional design questions. The conventional advice to developing democracies is that moderation and convergence promote economic stability. Our evidence suggests the opposite holds over most of the observed range: countries whose party systems offer clearer programmatic alternatives experience *less* growth volatility, not more, because the informational content of elections improves and within-government policy becomes more predictable.

The practical implication is that institutional reforms aimed at strengthening programmatic party competition may be more effective at reducing growth volatility than reforms aimed at ideological convergence. Policies that support party institutionalization, encourage internal discipline, and reward programmatic reputation building (such as public financing tied to policy platform development, or ballot structures that reward party labels over personal brands) could push developing democracies toward the stabilizing middle range of the polarization distribution. Conversely, reforms that weaken party organizations or encourage personalistic competition risk moving countries toward the low-polarization arm of the U, where elections convey little information and growth volatility rises.

Two caveats are important. First, the scope condition matters: the polarization channel operates most clearly when external shocks do not dominate. For commodity-dependent economies in periods of global turbulence, domestic political structure is a second-order determinant of growth volatility. Second, the relationship is nonlinear. The same reform that stabilizes growth by increasing polarization from low to moderate levels would destabilize growth if applied in an already highly polarized environment. Policy advice must be calibrated to the country's current position on the polarization distribution.

## 9. Conclusions

This paper has developed and tested a nonlinear theory of how party system polarization shapes the volatility of economic growth in developing democracies. The central finding is a U-shaped relationship: growth volatility falls as polarization increases from low to moderate levels, then rises

as polarization moves further. Moderate programmatic differentiation stabilizes growth; ideological convergence and extreme polarization both destabilize it, through distinct but theoretically related mechanisms.

The evidence is strongest in Latin America, where both the linear and squared terms of the Dalton vote-share polarization index are significant at the 1% level, with a turning point near 3.2 on a 0–10 scale. The lag structure supports a polarization-to-volatility causal direction, and the post-2007 attenuation confirms a scope condition about external shock dominance. The pattern replicates in non-Latin American democracies using a related polarization measure. A mechanism investigation testing nine candidate channels finds no sequential mediator: the effect operates through a simultaneous shift in the aggregate expectations environment, consistent with the variance decomposition theory.

These findings should be interpreted with appropriate caution given the observational identification strategy, the modest within-$R^2$ values, and the 16-country Latin American sample. But the consistency of the pattern across specifications, periods, and samples suggests it reflects a real feature of how democratic competition shapes economic outcomes in the developing world. Future work with richer measures of domestic policy uncertainty and longer panels would allow sharper tests of the proposed mechanism and more precise estimation of the turning point across institutional contexts.

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